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RESEARCH ARTICLE

Joint Signal-Power and Direction of Arrival Estimation for Far-Field Targets Using Novel Metaheuristic Framework

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ABSTRACT Accurate estimation of signal power and Direction of Arrival (DOA) from array data is a challenging problem with numerous applications in signal processing, radar, sonar and seismic exploration. Once the DOA information is gained, it is given to the beamformer network to place nulls in jammer directions, while maintaining the mainbeam in the desired look direction. Similarly, the estimated power is used to calculate the radar cross section that represent the effective reflecting area of the target. This work, proposes a unified framework based on the Artificial Protozoa Optimization (APO) Algorithm to jointly estimate the signal power and DOA of multiple far-field targets. A system model linking the estimated amplitudes to their corresponding power levels is formulated, and the joint estimation task is posed as a single dimensional search problem. A cost function based on Mean Square Error (MSE) is designed that defines the difference between actual and desired output of the system. This cost function neither require derivative information nor any strict statistical assumptions to explore the search space. It automatically pair the desired DOA with the relevant power corresponding to any particular target and does not require the multidimensional search. Simulation results verify that the proposed framework provides better estimation accuracy for both DOA and power of the far field targets even in the presence of low signal to noise ratio. The proposed framework is also compared with already existing algorithms in terms of estimation accuracy, MSE and robustness against noise.

INDEX TERMS Antenna arrays, artificial protozoa optimization, direction of arrival estimation, power estimation, radar, meta heuristic computing.

I. INTRODUCTION

Parameter estimation of signals impinging on an antenna array has been the hot research topic for several decades with direct applications in radar [1], seismic exploration [2], sonar [3], remote sensing [4] and medical imaging [5]. The key parameters of interest are Direction of Arrival (DOA) [6], signal power [7], range [8] and frequency [9]. By exploiting

the spatial diversity of antenna array, DOA information enables the beamforming network to positioned the nulls and main beam exactly in the jammer and required look directions respectively [10], [11], [12]. Over the last few decades, subspace based approaches, such as Multiple Signal Classification (MUSIC) and Estimation of Signal Parameter Through Rotational Invariant Techniques (ESPRIT) are widely used to estimate the DOA of signals impinging on antenna arrays [13], [14], [15], [16]. These algorithms exploit the eigen structure of the signal and noise subspaces and

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offer better angular resolution and robustness against noise. However, the performance of these algorithms tends to spoil for highly correlated signals and often requires a sufficient number of snapshots to achieve the desired results [17]. The MUSIC and ESPRIT algorithms for 1D DOA estimation can be extended for the 2-D parameters (Elevation & Azimuth angles) estimation, however their eigenvalue decomposition and heavy peak searching requirements does not make them suitable for estimating more than one parameter. In order to reduce the computational complexity, a 2D propagator method in [18] is proposed, which is also a sub space method but without the requirements of eigenvalue decomposition. In [19], another efficient technique called reduced dimension propagator method is developed for 2-D parameters that further reduces the computational cost. In the same way, [20] proposed another algorithm called 1D- ESPRI,T which is applied twice on the received signal, however this method is having the issue of pair matching. Another efficient double search technique is developed in [21] that simplifies the 2D search spectrum, however this method is also having the issue of pair matching between the corresponding parameters. Beside having the issues of computational complexity and pair matching, the above mentioned method does not estimate the power of the received signals along with the DOA. In many radar applications, the DOA information does not fully characterize the target and the received signal power information is also crucial, which is the main part of Radar Cross Section (RCS) linked with the target reflectivity [22], [23]. Estimating DOA along with received target power provide additional detection support, situational awareness, classification tasks and improve the reliability of downstream radar processing blocks [24], [25]. Despite the importance, joint DOA and power estimation have gained comparatively little consideration, as most existing techniques either estimate DOA and Power separately or rely on the simplified assumptions that do not hold in realistic environment [26], [27].

Recent development and evolution in metaheuristic optimization techniques have opened new avenues for researchers to solve complex nonlinear optimization problems [28]. Unlike traditional techniques, metaheuristic algorithms are gradient free and does not require the derivative of the problem. These techniques got further attention due to their ability in mitigating the local minima in the search space during the optimization process. Almost all the metaheuristic techniques starts with the vectors of a random population, where each vector is a possible candidate solution tested by a fitness function [29], [30]. The most widely used heuristic techniques are Particle Swarm Optimization [31], Genetic Algorithm [32], Cultural Algorithm [33], Differential evolution [34], Bee colony optimization [35] and Artificial Protozoa Optimization (APO) algorithm [36]. Currently the APO, which is inspired from the natural life process of protozoa, attracted the attention of research community to solve different problems arising in the field of science of technology [37]. As per the author information, no one has

developed and used the strategy based on APO for the joint estimation of power and DOA of far filed targets.

This work, proposes a unified framework based on APO Algorithm to jointly estimate the signal power and DOA of multiple far-field targets. A system model linking the estimated amplitudes to their corresponding power levels is formulated, and the joint estimation task is posed as a single search problem. A cost function based on Mean Square Error (MSE) is designed that works on the concept of error between desired and estimated output of the system. This cost function neither require derivative information nor any strict statistical assumptions to explore the search space. It automatically paired the desired DOA and power corresponding to any particular target. Simulation results verify that the proposed framework provides better estimation accuracy for both DOA and power of the far field targets impinging on uniform linear array even in the presence of low signal to noise ratio. The presented framework is also compared with already existing algorithms in terms of estimation accuracy, MSE and robustness against noise.

Some of the salient feature of the current work are as given below:

- A mathematical model based on DOA and power of far field targets is developed.
- A novel approach APO is designed and implemented to jointly estimate the DOA and Power of the received signal.
- A cost function based on the MSE is formulated and optimized that automatically paired the desired power and DOA corresponding to any particular target in the far field.
- The proposed algorithm is tested through extensive simulations for different scenario and compared with the already existed algorithms. The results shows that the proposed APA produced lower estimation error and better convergence across different scenarios.

The rest of the paper is organized as follows: Date model for field targets impinging on uniform Linear Array (ULA) is discussed in section II, while the proposed method based on APO is implemented for the joint estimation in section III. In the same way, the testing of the proposed APO is carried in section IV, while section V is dedicated for the conclusion and future work directions.

Throughout this work, matrices, vectors and scalars are represented by capital bold, small bold and small un-bold letters respectively.

II. SYSTEM MODEL FOR SIGNAL POWER AND DOA ESTIMATION

The target power and DOA estimation have the key role in radar signal processing. The received target power is one of the important parameter in finding the RCS. This information is crucial in determining the target size and reflectivity properties i.e., large received power means bigger size or higher reflective target. Similarly, DOA estimation gives the information about the target direction that can be used in

forming the beam in desired direction, while maintaining the nulls in the undesired directions. In this work, we assume a passive mono-static phase array radar whose receiver is equipped with uniform linear array (ULA) that has I sensors with same inter-element spacing d , as shown in Fig. 1. L narrow band targets are assumed in the far field that are independent from each other. The space around any antenna array can be broadly divided into three categories based on the distance of the target from that array. Mathematically these spaces can be defined as,

Reactive Near Field:

$$dis \tan ce < 0.62\sqrt{\frac{1}{\lambda}D^3}$$

Radiating Near Field (Fresnel):

$$0.62\sqrt{\frac{1}{\lambda}D^3} < dis \tan ce < \frac{2}{\lambda}D^2$$

Far field (Fraunhofer):

$$dis \tan ce \geq \frac{2}{\lambda}D^2$$

where D and λ represents the dimension of antenna and wavelength of the signal respectively. If the targets are lying in the far field, the impinging signals are considered to be plane waves as shown in fig. 1. The following plane wave assumptions are used when the target is located in the far field.

Amplitude Assumption: For any particular target impinging on ULA, the amplitudes remain the same i.e.,

$$a_1 \approx a_2 \approx \dots \approx a_L$$

DOA Assumption: Similarly, for any particular target, the DOA's are also remain the same i.e.,

$$\beta_1 \approx \beta_2 \approx \dots \approx \beta_L$$

Phase assumption: As the wavefront is planar, so the phase difference between the adjacent antennas is determined by the propagation delay.

In general, the output of the i -th antenna in the ULA for $L \leq I$ targets, can be formulated as:

$$x_i(t) = \sum_{l=1}^L a_l(t) \exp(-j((i-1)\psi_l)) + n_i(t) \quad (1)$$

where $i = 1, 2, \dots, I$. In equ. (1), $a_l(t)$ represent the amplitude of the l -th target (transmitter) and can be related to signal power by using the following relation,

$$p_l(t) = \frac{a_l^2(t)}{2} \quad (2)$$

where $l = 1, 2, \dots, L$. Using the definition of equ (2), we can re-write equ (1), as given below,

$$x_i(t) = \sum_{l=1}^L \sqrt{2P_l(t)} \exp(-j((i-1)\psi_l)) + n_i(t) \quad (3)$$

Similarly, in equ. (3), ψ_l defines the relative delay between the i -th and the reference antennas for the l -th target and can be given as:

$$\psi_l = kd \cos \beta_l \quad (4)$$

In equ. (4), the propagation constant $k = 2\pi/\lambda$ and the inter-element spacing $d = \lambda/2$, while β_l is the DOA of the l -th target. In the same way, n_i is the additive white gaussian noise added in the received signal at i -th antenna. It is assumed that the noise have zero mean and unit variance. For the entire L -targets, the response of the ULA in the vector form for single snapshot can be given as below:

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_I \end{pmatrix} = \begin{pmatrix} 1 & \dots & 1 \\ e^{-j\pi \cos \beta_1} & \dots & e^{-j\pi \cos \beta_L} \\ \vdots & \dots & \vdots \\ e^{-j\pi I \cos \beta_1} & \dots & e^{-j\pi I \cos \beta_L} \end{pmatrix} \begin{pmatrix} \sqrt{2p_1} \\ \sqrt{2p_2} \\ \vdots \\ \sqrt{2p_L} \end{pmatrix} + \begin{pmatrix} n_1 \\ n_2 \\ \vdots \\ n_I \end{pmatrix} \quad (5)$$

In more compact form equ. (5) can be represented as:

$$\mathbf{x} = \mathbf{D}\mathbf{p} + \mathbf{n} \quad (6)$$

In equ. (6), $\mathbf{D}$ is the array manifold matrix whose vectors contains the angle information related to the signal received from the far field targets and can also be given as:

$$\mathbf{D} = [\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_L, \dots, \mathbf{d}_L]^T \quad (7)$$

where $\mathbf{d}_l$ is called array manifold vector as defined in equ. (5). It is clear from the data model that the unknown parameters are the DOA $\{\beta_l\}_{l=1}^L$ and power $\{P_l\}_{l=1}^L$ of the received signals that needs to be estimated correctly and jointly.

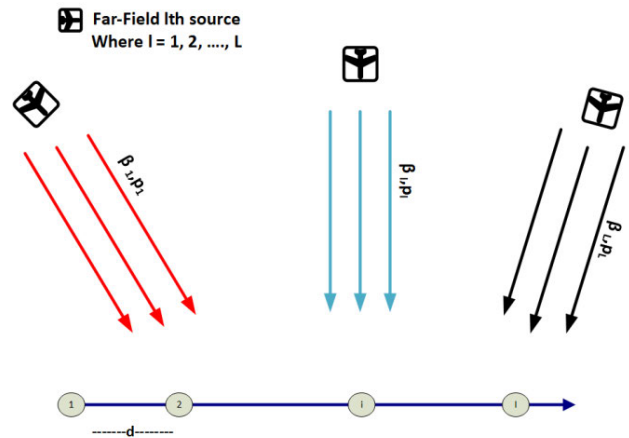


FIGURE 1. Signal model for far field targets.

III. PROPOSED METHODOLOGY

This section deal with implementation and designing of the proposed Artificial Protozoa Optimizer (APO) for the joint estimation of DOA and power of the far field targets. APO is metaheuristic population based algorithm inspired from the natural life process of Protozoa that include its survival strategy, reproduction process and foraging behavior in complex circumstances [36], [37]. The APO exploit the conduct of protozoa in searching the rich nutrient region and its reproduction process to get any global optimization efficiently.

Its formulation does not require the slope information and depends on limited number of controlled parameter. Due to its flexible structure, APO got wide range of applications arising in complex engineering problems [38], real world problems [39], energy management [40] etc. In the current work, APO is designed to efficiently estimate the power and DOA of far field targets. The flow diagram of APO is shown in Figure. 2 while its step in the form of pseudo code are given below:

Step i. Initialization: In this step, initial population is created randomly, where each vector called Protozoa is a possible candidate solution. Mathematically, a population contains K Protozoa can be given as:

$$\hat{\mathbf{V}} = \begin{pmatrix} \hat{\beta}_{11}\hat{P}_{11}, \hat{\beta}_{12}\hat{P}_{12}, \dots, \hat{\beta}_{1L}\hat{P}_{1L} \\ \hat{\beta}_{21}\hat{P}_{21}, \hat{\beta}_{22}\hat{P}_{22}, \dots, \hat{\beta}_{2L}\hat{P}_{2L} \\ \vdots \\ \hat{\beta}_{K1}\hat{P}_{K1}, \hat{\beta}_{K2}\hat{P}_{K2}, \dots, \hat{\beta}_{KL}\hat{P}_{KL} \\ \vdots \\ \hat{\beta}_{K1}\hat{P}_{K1}, \hat{\beta}_{K2}\hat{P}_{K2}, \dots, \hat{\beta}_{KL}\hat{P}_{KL} \end{pmatrix}^T \quad (8)$$

The upper and lower limit of β and P in each vector are $0 < \beta < \pi$ and $L_{\text{lim}} < P < U_{\text{lim}}$ respectively. Equ (8), can also be given as:

$$\hat{\mathbf{V}} = [\hat{\mathbf{v}}_1, \hat{\mathbf{v}}_2, \dots, \hat{\mathbf{v}}_k, \dots, \hat{\mathbf{v}}_K]^T \quad (9)$$

where each $\mathbf{v}$ represent a vector of protozoa in equ (8).

Step ii. Cost Function: In this work, cost function is designed by exploiting the concept of MSE. It can be given as follows:

$$MSE = \frac{1}{I} \left(E \left(\sum_{i=1}^I \sum_{k=1}^K |x_i - \hat{x}_i|^2 \right) \right) \quad (10)$$

where in equ. (10), x_i is given in equ. (3), while $\hat{x}_i$ is defined as follows:

$$\hat{x}_i = \sum_{k=1}^K \sqrt{2\hat{p}_k(t)} \exp(-j((i-1)\hat{\psi}_k)) \quad (11)$$

In equ. (10), x_i denotes the true parameter (signal power or DOA) and $\hat{x}_i$ represents its corresponding estimate. In equ. (11),

$$\hat{p}_k(t) = \frac{\hat{a}_k^2(t)}{2} \quad (12)$$

$$\hat{\psi}_k = \pi \cos \hat{\beta}_k \quad (13)$$

Step iii. Sorting and Nutrient Assignment: In this step, nutrient levels are assigned to each protozoa based on their fitness values. Protozoa with higher fitness (lower MSE) are assigned higher nutrient levels that control their movements and persistence during the optimization process.

Step iv. Foraging: A protozoa uses chloroplasts to produce carbohydrates to supply nutrition. Protozoa prefer a place to live that is less exposed to light, so it changes its position if that place is exposed to strong light intensity. The protozoan moves around the the qth protozoa if the light intensity is low around the k-th protozoa. It moves toward the current

best location by using small random perturbation. The new position is mathematically given as:

$$\mathbf{v}_k^{t+1} = \mathbf{v}_k^t + \delta\varphi(t) + (\mathbf{v}_{\text{best}}^t - \mathbf{v}_k^t)\gamma(t) \quad (14)$$

where $\gamma(t)$ and $\varphi(t)$ are adaptive step size parameters while δ is represent a gaussian random vector.

Step v. Swarming and interaction: This step deals with the enhancement of global exploration. The protozoa interacts with randomly chosen protozoa as given as:

$$\mathbf{v}_k^{t+1} = \mathbf{v}_k^t + \gamma(t)(\mathbf{g} - .5) + \varphi(t)(\mathbf{g} - .5) \odot (\mathbf{v}_k^t - \mathbf{v}_i^t) \quad (15)$$

where $\mathbf{g}$ is a uniformly distributed vector while $\mathbf{v}_i$ is a randomly chosen protozoa.

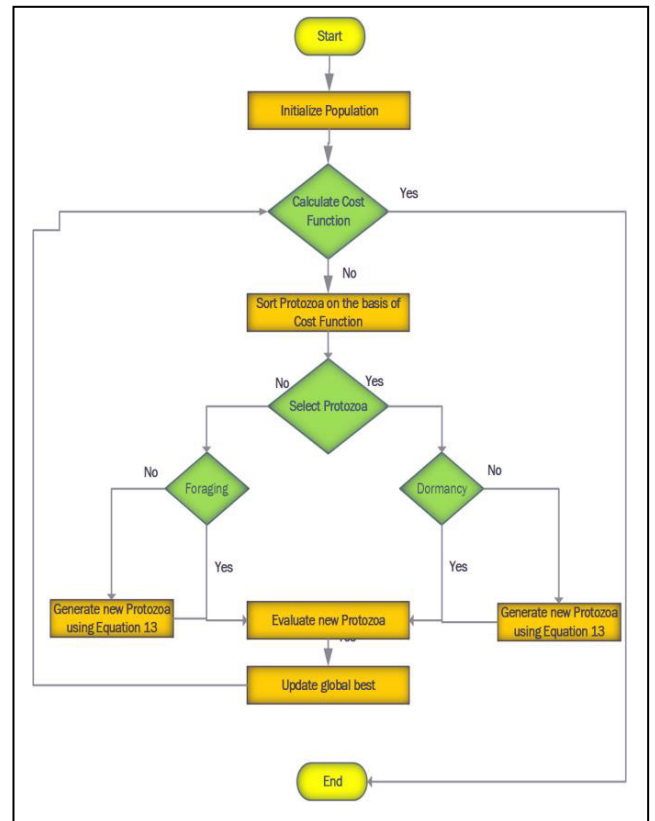


FIGURE 2. Flow chart for proposed APO.

Step Vi. Dormancy: This step tries to avoid getting stuck in the local minima and prevent any pre-mature convergence of the algorithm by reinitializing any weak protozoa during the optimization process as given below:

$$\mathbf{v}_i^{t+1} \sim U(\mathbf{v}_{\min}, \mathbf{v}_{\max}) \quad (16)$$

Step vii. Constraint Handling: In this step, all the unknown variables in the given problem are projected back in the defined region bounded by their upper and lower bounds.

Step viii. Global Best Update: The updated protozoa is accepted only if the fitness is improved.

Step ix. Termination: Based on the satisfaction of any of the following conditions, the algorithm will stop i.e.,

- The desired MSE is obtained
- Maximum iterations are completed

The final estimated values of DOA and power can be obtained from the best protozoa.

x. Storage: Store the best achieved results for later on analysis and discussion.

IV. RESULTS AND DISCUSSION

This section deals with the validation of the proposed APO algorithm for the joint estimation of signal power and DOA of far field targets. Throughout the simulations, the distance between two antenna in the ULA is kept same. All the DOA values are taken in degrees. Noise is considered to be AWGN with zero mean and unit variance. The initial population is based on 200 protozoa which are tested for 100 independent trials. The results are performed in MATLAB with system specifications as given in Table. 1.

TABLE 1. System specifications.

S.No	Item/product	Specification
1	Processor	Intel(R) Core(TM) i7-8565U CPU @ 1.80GHz 1.99 GHz
2	Installed RAM	48.0 GB
3	SSD	477 GB SSD SAMSUNG MZVLB512HBJQ-000L7
4	Graphics Card	NVIDIA Quadro P520 (2 GB), Intel(R) UHD Graphics 620 (128 M)
5	System Type	64-bit operating system, x64-based processor
6	MATLAB	R2023b

Estimation accuracy: This section is dedicated to validate the estimation accuracy of the APO for two, three and four far field targets in the presence of noise. For each case, the ULA is having twice the number of antennas to that of the number of targets. In Table 2-4, the APO is tested for two, three and four targets in the presence of 10dB noise. The result shows that the estimation accuracy is very good especially for 2 targets, while bit degraded for 3 and 4 targets but still the results are well accepted. Similarly, Table 5-7, provides the estimation accuracy for 2, 3 and 4 targets in the presence of 5dB. Again one can see that even in low SNR the estimation accuracy of the APO is up to the mark.

TABLE 2. Estimation accuracy for 2 targets with 10 dB noise.

Targets	Desired DOA	Desired Power	Estimated DOA	Estimated Power
Target1	35	0.5	35.0003	0.5012
Target2	65	4.5	64.9998	4.5025

In Table. 8, the performance of proposed algorithm is carried out in terms of proximity effect in the presence of even low SNR condition. Again, the results demonstrate that the APO produced fair good enough results when the targets are located close to each other that further validate its effectiveness.

TABLE 3. Estimation accuracy for 3 targets with 10 dB noise.

Targets	Desired DOA	Desired Power	Estimated DOA	Estimated Power
Target1	36	1.1250	36.0019	1.1282
Target2	66	6.8450	66.0031	6.8345
Target3	76	12.0050	75.9997	12.0072

TABLE 4. Estimation accuracy for 4 targets with 10 dB noise.

Targets	Desired DOA	Desired Power	Estimated DOA	Estimated Power
Target1	37	2.0000	36.9988	1.9967
Target2	57	4.8050	56.9979	4.8033
Target3	67	10.1250	66.9994	10.1203
Target4	77	16.2450	77.5526	16.2416

TABLE 5. Estimation accuracy for 2 targets with 5 dB noise.

Targets	Desired DOA	Desired Power	Estimated DOA	Estimated Power
Target1	50	4.5000	49.01213	4.5091
Target2	110	2.6450	110.0452	2.6501

TABLE 6. Estimation accuracy for 3 targets with 5 dB noise.

Targets	Desired DOA	Desired Power	Estimated DOA	Estimated Power
Target1	40	8.0000	40.4001	8.3282
Target2	70	0.2450	69.5998	0.3001
Target3	120	15.1250	120.4120	15.5007

TABLE 7. Estimation accuracy for 4 targets with 5 dB noise.

Targets	Desired DOA	Desired Power	Estimated DOA	Estimated Power
Target1	35	4.5000	35.8786	4.9967
Target2	60	1.4450	59.2999	1.8033
Target3	105	10.1250	104.2101	10.7803
Target4	77	0.5000	77.7026	0.7416

TABLE 8. Proximity effect 3 targets with -5 dB noise.

Targets	Desired DOA	Estimated DOA	Desired DOA	Estimated DOA
Target1	40	40.6101	67	67.7321
Target2	70	69.3763	70	70.6298
Target3	73	73.7120	73	72.2620

MSE: In this sub-section, the performance of APO in terms of MSE is described. In figures 3 and 4, the performance is carried out for 2, 3 and 4 targets in the presence of 10dB and 5dB noises respectively. In both the cases, the results are fairly well enough against increasing number of runs. In the same way, the Root Mean Square Error (RMSE) is validated against the different values of SNR for 2, 3 and 4 targets as

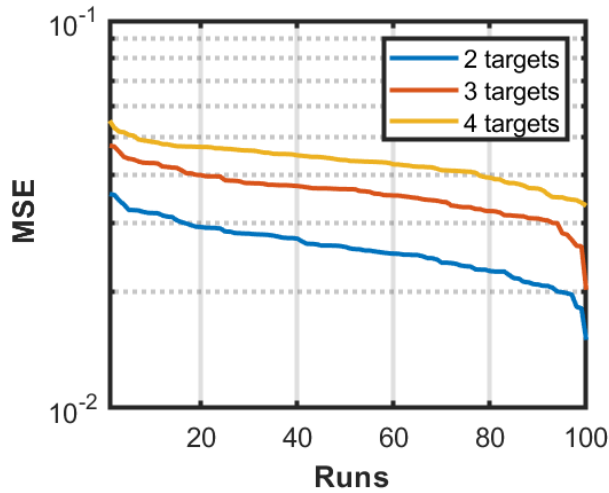


FIGURE 3. MSE performance in the presence of 10 DB noise.

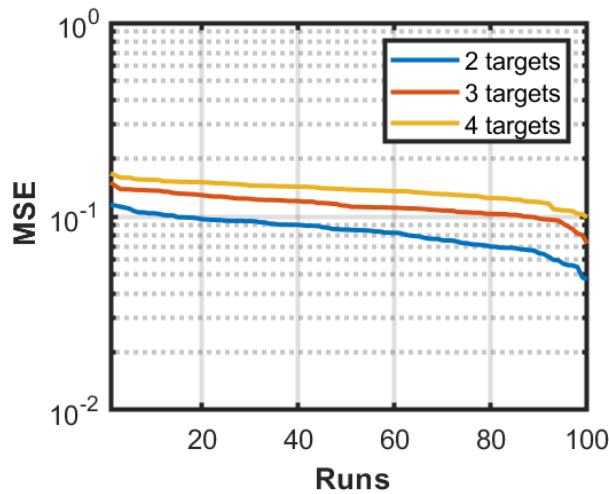


FIGURE 4. MSE performance in the presence of 5 DB noise.

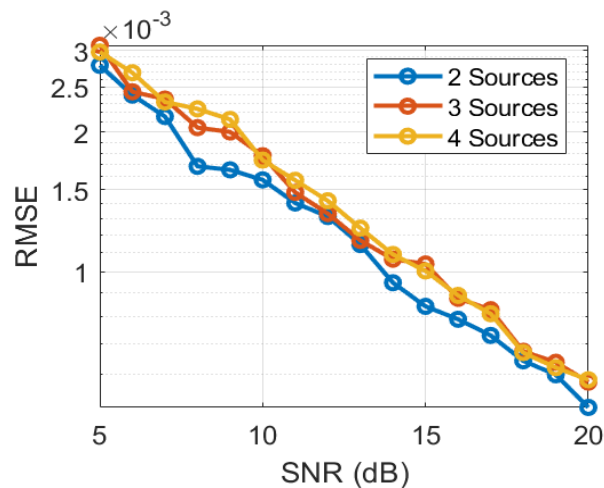


FIGURE 5. RMSE against SNR for 2, 3 and 4 targets.

shown in figure 5. The robustness of APO against noise is excellent.

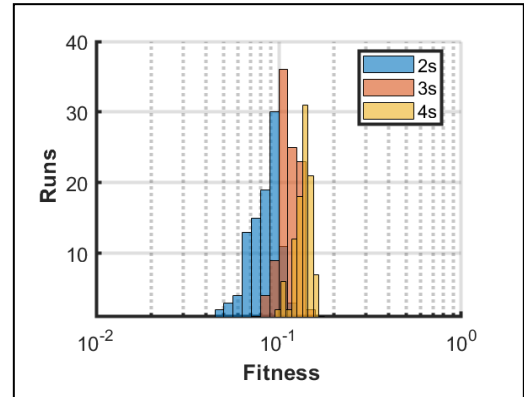


FIGURE 6. Histogram analysis for 2,3 and 4 targets with 05 dB noise.

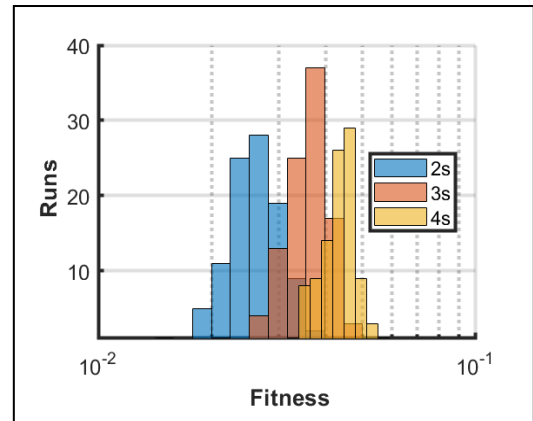


FIGURE 7. HISTOGRAM analysis for 2,3 and 4 targets with 10 DB noise.

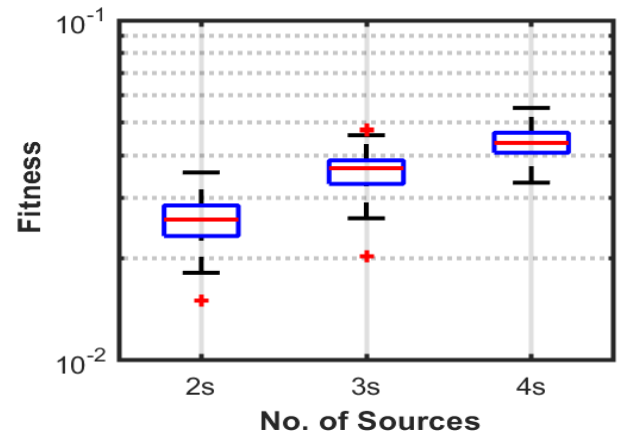


FIGURE 8. BOX plot for fitness vs targets with 10 DB noise.

Figures 6 and 7 validated the efficiency of APO through histogram plots for 2, 3 and 4 targets in the presence of 5 dB and 10 dB noises respectively. Almost 30 runs are required for 2 targets to get a fitness value around 10^{-2} and so on.

In Figures 8 and 9, the performance of APO is validated through box plots for 2, 3 and 4 targets in the presence of 10 dB and 5 dB noise respectively. It is clear that the APO achieved better fitness against each number of targets even in the presence of low SNR.

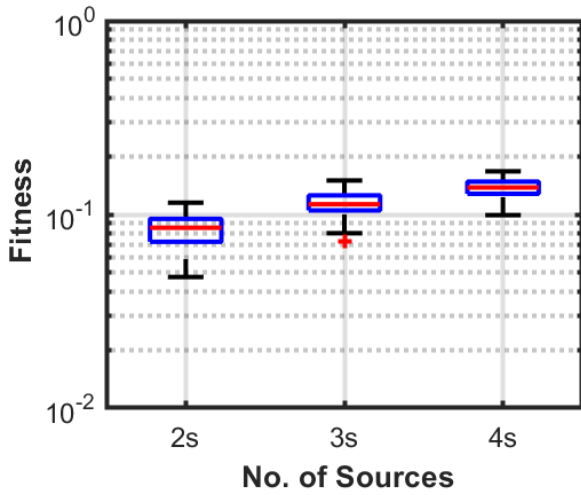


FIGURE 9. BOX plot for fitness vs no of targets with 5 DB noise.

In figures 10 and 11, the performance of APO is evaluated in terms of cdf plots against the fitness values for 2, 3 and 4 targets in the presence of 10 db and 5 db noise respectively. Again the result shows that APO achieved excellent results in each case.

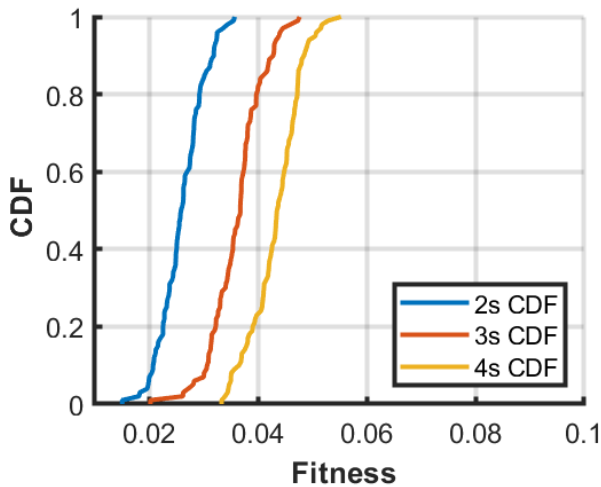


FIGURE 10. CDF vs fitness for different targets with 10 DB noise.

Comparison with state of the Art Algorithm: In this section, the estimation accuracy and computational complexity of APO is compared with the other Meta Heuristic technique called Flower Pollination Algorithm (FPA). The FPA is used in [41] to estimate the DOA only of the far field targets, while it does not estimate the power. Thus, the comparison of APO is mainly performed with FPA for DOA estimation only. In order to get fair comparison, a population size of 200 candidate solutions are used for both the algorithms. Similarly, both the algorithms are executed for 100 independent trials and their stopping criteria depend on any of the condition is achieved as given in step ix of the proposed methodology.

In this case, we considered three targets whose desired angles are 70° , 110° and 40° , while 5 dB noise is added

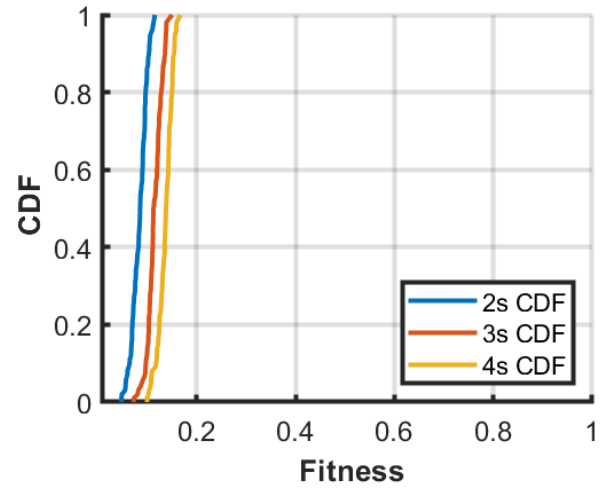


FIGURE 11. CDF vs fitness for different targets with 05 dB noise.

to the system. As provided in Table. 8, T, T1, T2 and T3 represent targets, target 1, target 2 and target 3 respectively. It is established that the estimation accuracy of the APO is significantly better than that of the FPA. In the same way, the computational complexity of APA is better than that of FPA as it requires 0.4173 seconds to achieve the required results. Both the proposed APO and the benchmark FPA are population based meta heuristic optimization techniques, where the main memory is occupied by storing the population of the possible candidate solutions. For example, the population contains K Protozoa where the dimension of each protozoa is $2 \times L$. L represent the total number of impinging targets on ULA, where each target correspond to two unknown parameters called signal Power and Direction of Arrival (DOA). Therefore the space complexity can be given to be $O(K \times 2L)$. In the given simulation, we have considered three targets with 200 protozoa, where the dimension of each protozoa is $2 \times 3 = 6$. In this regard, the total memory required for the storage of population is 1200 floating time number, while extra 200 numbers are needed for storing fitness values of each protozoa that makes total number to be 1400. Therefore, the final total memory requirements is approximately $1400 \times 8 = 11200$ bytes roughly equal to 11KB. Therefore, both the algorithms bears comparably the same space complexity. However, the proposed APO attained significantly lower computational time complexity $O(T \times K \times 2L)$ due to its first convergence because it needs less iterations (T) to converge. As discussed in Table. 9, for three targets 0.4173 seconds are required for APO, while FPA needs 1.5297 seconds to achieve the final results.

TABLE 9. Comparison analysis of APO and FPA with 5 dB noise.

T	Desired DOA	FPA Estimation	APO Estimation	FPA Time(s)	APO Time(s)
T1	110	111.361	110.412	1.5297	0.4173
T2	70	71.599	69.599		
T3	40	38.7859	39.5998		

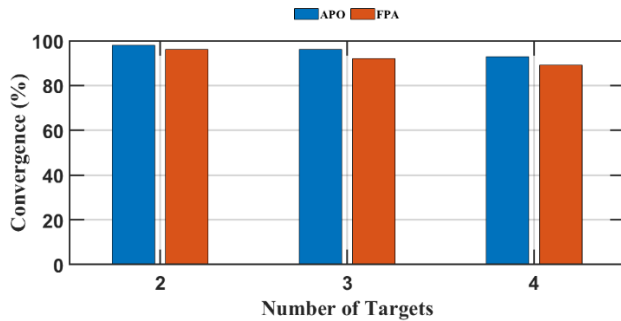


FIGURE 12. Convergence comparison of APO vs FPA with 05 dB noise.

Further, the convergence behaviour of APO can be understood through the results as shown in figure. 12. In this simulation, both the algorithms were run for 100 times and we observed the total number of times each algorithm achieved the accurate results for two, three and four targets respectively. As the results shows, APO consistently converges towards the desired parameter and achieved better convergence rate of 98%, 96% and 93% for two, three and four targets respectively as compared to FPA that attained 96%, 92% and 89% for the same scenario.

V. CONCLUSION AND FUTURE DIRECTIONS

In this work, joint DOA and power of far field targets were estimated using a novel approach called Artificial Protozoa Optimization (APO). A mathematical model that linked the targets power to the amplitudes of the impinging signals on ULA from different directions. The Power along with DOA was not addressed before using the APO algorithm, which is an important parameter that provides useful information about the size of a target. A fitness function based on MSE was designed and optimized using APO. The performance of APO was carried out in terms of estimation accuracy, robustness against noise and MSE. It was also compared with the other meta heuristic technique called FPA and it has been shown that APO performed better than FPA especially in low SNR environment.

In future, one can implement the same approach for bi-static MIMO radar where direction of departure should be estimated along with Direction of arrival. In the same way, the proposed algorithm can be implemented null steering and adaptive beamforming using different array geometries.

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REFERENCES

- [1] F. Zaman, S. Akbar, Z. Akhtar, and G. Phanomchoeng, "Efficient technique for joint parameter estimation of mixed far field targets and near field jammers in phased array radar," *IEEE Access*, vol. 13, pp. 55887–55898, 2025.
- [2] A. Almehdhar, A. Hamida, K. Aliyu, S. Alawsh, A. Muqabel, S. Al-Dharrah, W. Mesbah, and G. L. Stuber, "DOA estimation in wireless seismic surveys using deep learning," in *Proc. Int. Conf. Commun., Signal Process., Appl. (ICCSA)*, Mar. 2021, pp. 1–6.
- [3] N. Ahmed, H. Wang, M. A. Z. Raja, W. Ali, M. Kashif, and F. Zaman, "Novel design of grey wolf optimization heuristics for high resolution direction of arrival estimation in acoustic plane waves," *Wireless Pers. Commun.*, vol. 128, no. 4, pp. 2507–2529, Feb. 2023.
- [4] Q. Shen, W. Liu, W. Cui, and S. Wu, "Underdetermined DOA estimation under the compressive sensing framework: A review," *IEEE Access*, vol. 4, pp. 8865–8878, 2016.
- [5] N. Dey, A. S. Ashour, F. Shi, and R. S. Sherratt, "Wireless capsule gastrointestinal endoscopy: Direction-of-arrival estimation based localization survey," *IEEE Rev. Biomed. Eng.*, vol. 10, pp. 2–11, 2017.
- [6] F. Zaman, A. Iqbal, B. Ali, and A. K. J. Saudagar, "Joint estimation of elevation and azimuth angles with triple-parallel ULAs using Metaheuristic and direct search methods," *Comput. Model. Eng. Sci.*, vol. 145, no. 2, pp. 2535–2550, 2025.
- [7] H. Sun, L. Zhu, W. Mei, and R. Zhang, "Power-measurement-based channel autocorrelation estimation for IRS-assisted wideband communications," *IEEE Trans. Wireless Commun.*, vol. 24, no. 6, pp. 4647–4662, Jun. 2025.
- [8] A. El-Awamry, F. Zheng, T. Kaiser, and M. Khaliel, "Impact of phase noise on range estimation accuracy in harmonic FMCW radar systems," *IEEE Access*, vol. 13, pp. 42669–42688, 2025.
- [9] V. Montenegro, S. Dornetti, A. Ferraro, and M. G. A. Paris, "Enhanced quantum frequency estimation by nonlinear scrambling," *Phys. Rev. Lett.*, vol. 135, no. 3, Jul. 2025, Art. no. 030802.
- [10] C. M. Andras, G. Barb, and M. Otesteanu, "Comparative analysis of beamforming techniques and beam management in 5G communication systems," *Sensors*, vol. 25, no. 15, p. 4619, Jul. 2025.
- [11] J. Li and P. Stoica, *Robust Adaptive Beamforming*. Hoboken, NJ, USA: Wiley, 2005.
- [12] F. Zaman, "Estimation of direction of arrival for adaptive beamforming," Ph.D. dissertation, Dept. Electron. Eng., Fac. Eng. & Technol., Int. Islamic Univ. Islamabad, Pakistan, 2013.
- [13] A. Arshaghi, "Enhancing DOA estimation: MUSIC algorithm and comparative analysis of TOA, TDOA, and AOA techniques," *Int. J. Smart Electr. Eng.*, vol. 14, no. 1, pp. 25–31, 2025.
- [14] Y. Zhang and B. P. Ng, "MUSIC-like DOA estimation without estimating the number of sources," *IEEE Trans. Signal Process.*, vol. 58, no. 3, pp. 1668–1676, Mar. 2010.
- [15] Y.-M. Ning, S. Ma, F.-Y. Meng, and Q. Wu, "DOA estimation based on ESPRIT algorithm method for frequency scanning LWA," *IEEE Commun. Lett.*, vol. 24, no. 7, pp. 1441–1445, Jul. 2020.
- [16] J. Li, G. Yang, X. Ren, Z. Fu, and S. Qiu, "Modified ESPRIT and unitary-transformation form for computationally efficient DOA estimation on FPGA," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–9, 2025.
- [17] H. Hassan, M. Amin, and A. R. Maud, "Performance comparison of subspace and sparsebased DOA estimation algorithms," in *Proc. IEEE Int. Symp. Phased Array Syst. Technol. (ARRAY)*, Oct. 2024, pp. 1–5.
- [18] N. Tayem and H. M. Kwon, "L-shape two-dimensional arrival angle estimation with propagator method," *IEEE Trans. Antennas Propag.*, vol. 53, no. 5, pp. 1622–1630, May 2005.
- [19] M. Zhou, X. Zhang, X. Qiu, and C. Wang, "Two-dimensional DOA estimation for uniform rectangular array using reduced-dimension propagator method," *Int. J. Antennas Propag.*, vol. 2015, pp. 1–10, Aug. 2015.
- [20] Y.-Y. Wang and S.-C. Huang, "An ESPRIT-based algorithm for two-dimensional DOA estimation," *IEICE Trans. Fundamentals Electron., Commun. Comput. Sci.*, vol. 94, no. 9, pp. 1847–1850, Sep. 2011.
- [21] Z. Ye and C. Liu, "Two-dimensional DOA estimation in the presence of mutual coupling," *IEEE Trans. Antennas Propag.*, vol. 56, no. 10, pp. 3150–3158, Oct. 2008.
- [22] E. F. Knott, *Radar Cross Section Measurements*. New York, NY, USA: Springer, 2012.
- [23] A. A. Salama, "Direction of arrival estimation: A tutorial survey of classical and modern methods," 2025, *arXiv:2508.11675*.
- [24] S. Kulkarni, A. Thakur, S. Soni, A. Hiwale, M. H. Belsare, and A. A. B. Raj, "A comprehensive review of direction of arrival (DoA) estimation techniques and algorithms," *J. Electron. Electr. Eng.*, vol. 4, no. 1, pp. 138–186, 2025.
- [25] N. Ruan, H. Wang, F. Wen, and J. Shi, "DOA estimation in B5G/6G: Trends and challenges," *Sensors*, vol. 22, no. 14, p. 5125, Jul. 2022.

- [26] A. M. Molaei, B. Zakeri, S. M. H. Andargoli, M. A. B. Abbasi, V. Fusco, and O. Yurduseven, "A comprehensive review of direction-of-arrival estimation and localization approaches in mixed-field sources scenario," *IEEE Access*, vol. 12, pp. 65883–65918, 2024.
- [27] A. Mehmood, N. I. Chaudhary, A. Zameer, and M. A. Z. Raja, "Novel computing paradigms for parameter estimation in power signal models," *Neural Comput. Appl.*, vol. 32, no. 10, pp. 6253–6282, 2019.
- [28] X.-S. Yang, *Engineering Optimization: An Introduction With Metaheuristic Applications*. Hoboken, NJ, USA: Wiley, 2010.
- [29] C.-W. Tsai and M.-C. Chiang, *Handbook of Metaheuristic Algorithms: From Fundamental Theories to Advanced Applications*. Amsterdam, The Netherlands: Elsevier, 2023.
- [30] D. Oliva, M. A. Elaziz, and S. Hinojosa, *Metaheuristic Algorithms for Image Segmentation: Theory and Applications*. Cham, Switzerland: Springer International Publishing, 2019.
- [31] D. Zhu, R. Li, Y. Zheng, C. Zhou, T. Li, and S. Cheng, "Cumulative major advances in particle swarm optimization from 2018 to the present: Variants, analysis and applications," *Arch. Comput. Methods Eng.*, vol. 32, no. 3, pp. 1571–1595, Apr. 2025.
- [32] M. S. Hasan, M. Tarequzzaman, M. Moznuzzaman, and M. A. Ahad Juel, "Prediction of energy consumption in four sectors using support vector regression optimized with genetic algorithm," *Heliyon*, vol. 11, no. 2, Jan. 2025, Art. no. e41765.
- [33] S. U. Khan, I. M. Qureshi, F. Zaman, B. Shoaib, A. Naveed, and A. Basit, "Correction of faulty sensors in phased array radars using symmetrical sensor failure technique and cultural algorithm with differential evolution," *Scientific World J.*, vol. 2014, pp. 1–10, Aug. 2014.
- [34] F. Zaman, S. U. Khan, K. Ashraf, and I. M. Qureshi, "An application of hybrid differential evolution to 3-D near field source localization," in *Proc. 11th Int. Bhurban Conf. Appl. Sci. Technol.*, Jan. 2014, pp. 474–477.
- [35] S. Kumari and S. K. Singh, "An artificial bee colony optimized," in *Proc. 7th Int. Conf. Inf. Syst. Manage. Sci. AI Technol. Inf. Syst. Manage. Sci.*, vol. 2025, 2025, p. 420.
- [36] X. Wang, V. Snásel, S. Mirjalili, J.-S. Pan, L. Kong, and H. A. Shehadeh, "Artificial protozoa optimizer (APO): A novel bio-inspired Metaheuristic algorithm for engineering optimization," *Knowl.-Based Syst.*, vol. 295, Jul. 2024, Art. no. 111737.
- [37] X. Wang, V. Snásel, S. Mirjalili, and J.-S. Pan, "MAAPO: An innovative membrane algorithm based on artificial protozoa optimizer for multilevel threshold image segmentation," *Artif. Intell. Rev.*, vol. 58, no. 10, Jul. 2025, Art. no. 324.
- [38] M. Shehab, "Artificial protozoa optimizer: A bio-inspired Metaheuristic for complex engineering optimization," *Results Eng.*, vol. 27, Sep. 2025, Art. no. 106883.
- [39] H. Bo, J. Wu, and G. Hu, "MSAPO: A multi-strategy fusion artificial protozoa optimizer for solving real-world problems," *Mathematics*, vol. 13, no. 17, p. 2888, Sep. 2025.
- [40] R. G. Mohamed, A. A. Hassan, and S. H. E. Abdel Aleem, "A modified energy management strategy for PV/diesel hybrid system to reduce diesel consumption based on artificial protozoa optimizer," *Sci. Rep.*, vol. 15, no. 1, Feb. 2025, Art. no. 4440.
- [41] S. Akbar, M. A. Z. Raja, N. I. Chaudhary, F. Zaman, and H. Alquhayz, "Flower pollination heuristics for parameter estimation of electromagnetic plane waves," *Comput., Mater. Continua*, vol. 68, no. 2, pp. 2529–2543, 2021.



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